

BI Portfolio Project

Analytics Foundation for Google Merchandise Store

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Python · BigQuery · SQL · SEMRush · GA4 · DataStudio · Claude API



1. Project Overview

This project is a full-stack marketing intelligence dashboard built using GA4 ecommerce data from the Google Merchandise Store, enriched with SEMrush search engine data. This solution is powered by a BigQuery data warehouse designed around a dimensional star schema architecture, with layered transformation and reporting models that separate raw ingestion, business logic, and presentation layers.

The dashboard consists of three stakeholder-oriented pages, each tailored to a distinct decision-making context: acquisition performance monitoring for marketing leadership, SEO and landing page analysis for content and search teams, and competitive benchmarking for strategic planning.

Beyond traditional reporting and data visualization, the project incorporates an AI-powered insights layer. Channel performance metrics are automatically aggregated, analyzed through the Claude API, and written back into BigQuery as dynamically generated executive recommendations surfaced directly within the dashboard.

The objective of the project was to design a scalable BI and analytics solution capable of integrating multi-source marketing data into stakeholder-ready dashboards and actionable business insights. The project aims to demonstrate:

- Business-focused KPI framework and dashboard design
- Multi-source data integration, transformation, and validation
- Dimensional modeling and semantic reporting architecture
- SQL and BigQuery analytics engineering
- Stakeholder-oriented dashboard development
- Automated insight generation using Python and external APIs

Key Links

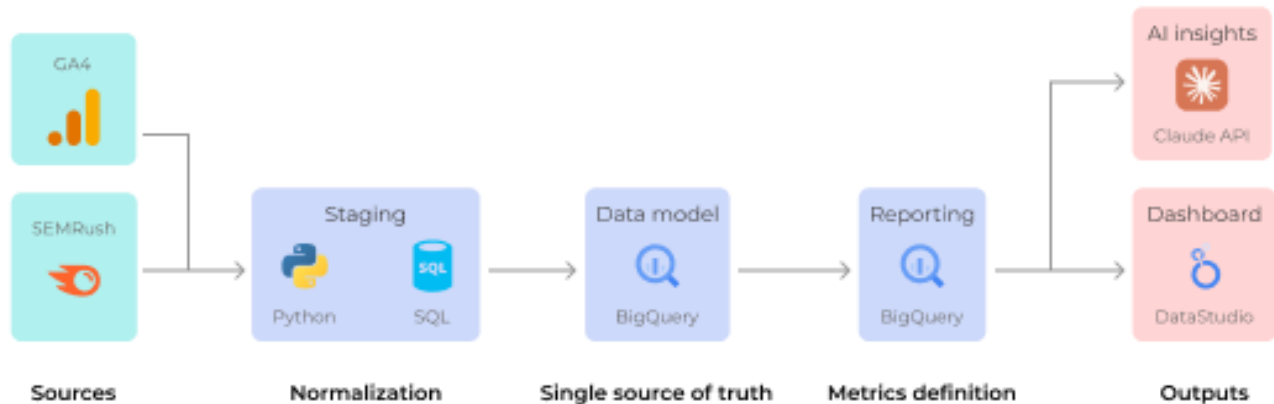
GitHub Repository

https://github.com/mdorego/BI_portfolio_project

Looker Studio Dashboard

<https://datastudio.google.com/reporting/1fce71dd-5c5d-439e-b095-4dfd8e6c5332>

2. Architecture & Data Stack



The dashboard is built on a layered BI architecture that separates ingestion, transformation, and delivery into distinct, independently maintainable stages. Raw source data is normalized in a staging layer, modeled into a star schema in the marts layer, and exposed through semantic reporting views consumed directly by Looker Studio. This layering decouples transformation logic, business metric definitions, and visualization.

A star schema was chosen over a flat or wide-table approach to centralize metric definitions, reduce redundancy across reporting views, and keep the model extensible as new dimensions or data sources are introduced.

The stack was selected for its tight native integration across the Google Cloud ecosystem and its widespread adoption in marketing analytics contexts:

- **Google BigQuery** serves as the core data warehouse and transformation layer. All staging and marts logic is written in SQL and executed via the BigQuery API, with Python orchestrating job submission, dependency ordering, and error handling. This keeps transformations version-controlled, auditable, and independent from the visualization layer.
- **Python** handles all orchestration outside of pure SQL logic: submitting transformation jobs to BigQuery in dependency order, ingesting SEMrush data, performing multi-step URL normalization for cross-source joins, and generating AI-powered insights through the Claude API.
- **SEMrush** provides the competitive intelligence layer through organic keyword rankings, traffic estimates, and domain-level benchmarks for merch.google and its three main competitors. This external data source enables the strategic dimension of the dashboard that GA4 alone cannot provide.

- **Looker Studio** acts as the visualization and delivery layer, connecting directly to BigQuery reporting views through live queries. The dashboard is interactive, supporting cross-page filters and drill-downs that allow stakeholders to explore the data across channels, devices, and time periods. An AI-generated recommendation card is included on the acquisition performance page as a demonstration of the intended insight delivery pattern; in a production setup this would refresh automatically on each pipeline run.
- **Claude API (Anthropic)** powers the automated executive insight card on the acquisition performance page. Channel performance metrics are passed as structured input, and the model returns a concise recommendation summary that is written back into BigQuery and surfaced directly in the dashboard.

3. Data Sources

The project combines two complementary data sources;

GA4 provides behavioral depth through session-level data covering traffic, engagement, conversion, and revenue across acquisition channels.

SEMrush adds competitive and keyword context through organic rankings, search volume estimates, and domain-level traffic benchmarks.

Combining both sources at the landing page level enables SEO analysis by correlating ranking position with actual conversion outcomes, while giving competitive benchmarking stronger business context beyond raw keyword counts.

Google Analytics 4 — Google Merchandise Store

The primary source is Google's publicly available GA4 BigQuery export for the Google Merchandise Store, covering November 2020 to January 2021. Working directly with raw event-level data rather than aggregated GA4 reports was a deliberate design choice, enabling session-level metric construction, custom engagement logic, and full control over business definitions such as landing page and channel grouping.

The staging layer reconstructs sessions from raw event rows, aggregates engagement and conversion metrics, and extracts each session's landing page from its first page-view event.

SEMrush — Organic & Competitive Keyword Intelligence

Keyword and competitive data was collected through the SEMrush API for four domains: merch.google, redbubble.com, zazzle.com, and cafepress.com. The dataset includes organic keyword rankings, estimated traffic value, and domain-level visibility metrics used for SEO benchmarking and competitor analysis.

Joining SEMrush data to GA4 sessions required resolving a structural mismatch: the GA4 export uses legacy URL conventions from an older domain structure, while SEMrush tracks the current

merch.google site architecture. This challenge was addressed through a multi-step URL normalization and categorization pipeline applied consistently across both the Python ingestion layer and SQL staging layer.

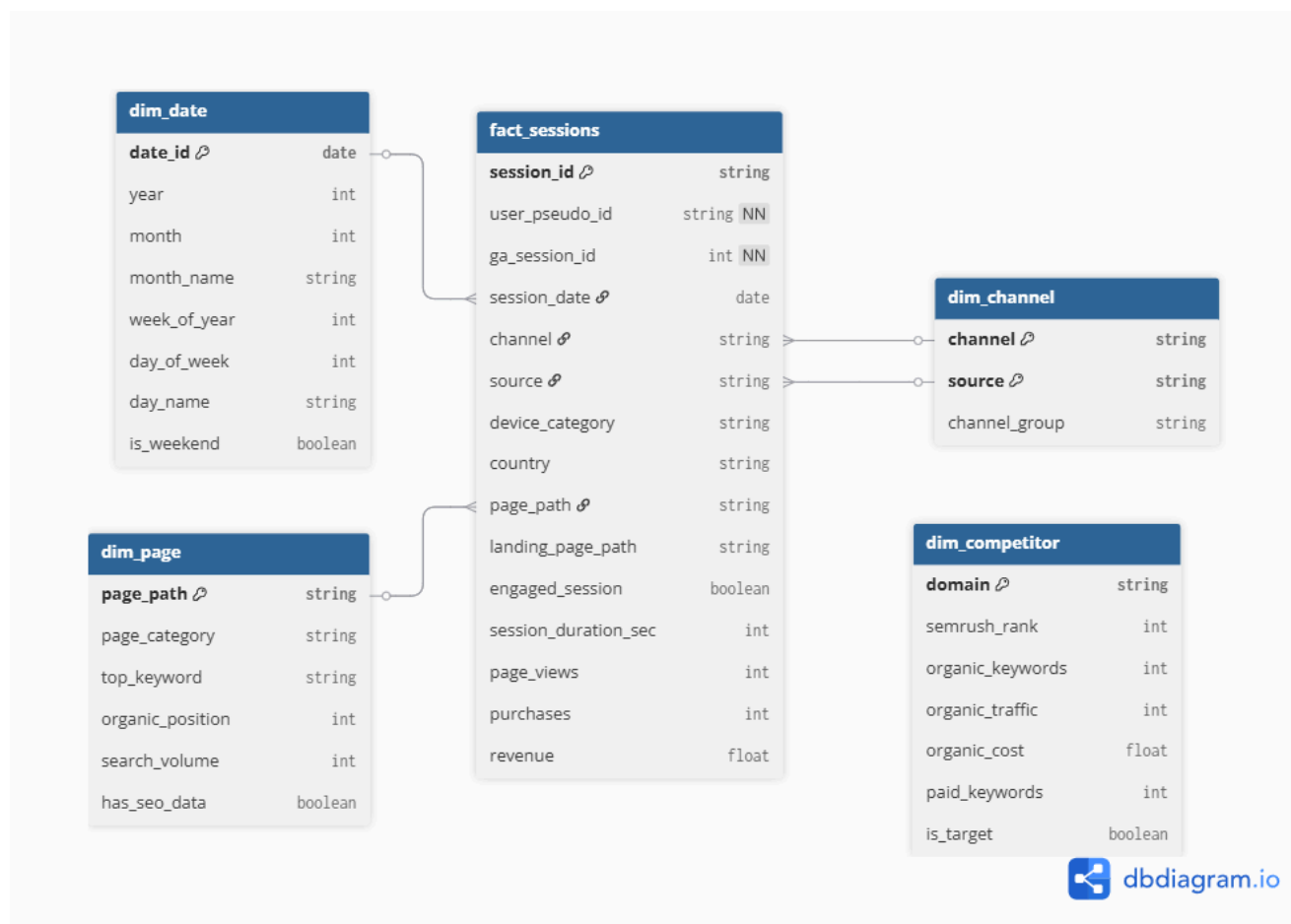
See Section 6 for implementation details, coverage rates, and remaining limitations.

4. Data Model

The data warehouse of this project is structured in three layers: a staging layer that cleans and normalizes raw source data, a marts layer containing the star schema that serves as the single source of truth for business metrics, and a reporting layer of pre-aggregated views consumed directly by Looker Studio. This layered structure is what enables the dashboard: Looker Studio connects exclusively to the reporting views, with all transformation and metric logic handled upstream in BigQuery.

Where possible, key metrics such as conversion rate and revenue per session are defined in the reporting layer rather than inside individual dashboard charts, keeping definitions consistent and easier to maintain.

Star Schema & Dimensions



The star schema centers on a session-level fact table, *fact_sessions*, joined to four dimension tables: *dim_date*, *dim_channel*, *dim_page*, and *dim_competitor*.

dim_channel applies Google's standard channel grouping logic, ensuring the model aligns with GA4's native acquisition reporting. *dim_page* is the cross-source join point between GA4 and SEMrush: it carries both behavioral attributes from GA4 and keyword attributes from SEMrush, with a boolean flag that explicitly marks pages where SEO data was available. This prevents sessions from being silently excluded when no SEMrush match exists.

Device type and country are kept directly on *fact_sessions* rather than in separate dimensions. Both have low cardinality and serve primarily as filters, so dedicated dimension tables would add join overhead without analytical benefit.

Field Glossary

engaged_session — Boolean set by GA4 when a session lasts more than 10 seconds, contains at least two page views, or includes a purchase. Used to compute the engagement rate KPI.

purchases — Count of purchase events within a session. Conversion rate treats this as binary, meaning sessions with multiple purchases still count as a single conversion.

has_seo_data — Indicates whether the landing page successfully joined to SEMrush keyword data. When false, ranking position and search volume fields remain null.

is_target — Identifies *merch.google* as the primary domain, distinguishing it from competitor rows stored in the same dimension.

organic_cost — SEMrush estimate of the monthly paid-search equivalent value of a domain's organic traffic. Surfaced as the Estimated Traffic Value metric.

Reporting Layer



The reporting layer exposes four SQL views that connect directly to Looker Studio. It is built in two stages: a wide session-level base view, **rpt_sessions_enriched**, joins all dimensions and pre-computes engagement and conversion flags, giving the three aggregated views above it a consistent, enriched starting point.

Those three aggregated views each map to one dashboard page: channel and device performance for the acquisition page, landing page behavioral and SEO metrics for the SEO page, and keyword ranking analysis for the competitive intelligence page.

Defining these metrics in SQL views rather than as Looker Studio calculated fields keeps the business logic version-controlled, reusable across reports, and independent from the visualization layer.

AI Insights Table



ai_insights	
generated_at	timestamp NN
insight_type	string NN
insight_text	string NN
created_at	timestamp NN

The AI insights table stores the output of each Claude API run: the recommendation text, a type field for categorization, and two timestamps — one recording when the API call was made (used by Looker Studio to surface the most recent recommendation), and one recording when the row was written to BigQuery (used for pipeline monitoring). The table follows an append-only design, allowing recommendations to be tracked historically as the underlying performance data evolves

5. Deliverable

Dashboard

The final deliverable is a three-page interactive Looker Studio dashboard designed around distinct stakeholder workflows. All pages share a global date range filter and device selector, with the default view comparing January 2021 performance against December 2020.

Page 1 — Traffic & Channels Performance

Business question: Which acquisition channels drive volume, and which convert and generate revenue efficiently?

The page is organized around two analytical perspectives:

- **Traffic generation:** Which channels drive the most sessions and engagement?
- **Revenue efficiency:** Which channels generate the highest conversion and revenue value?

Key visualizations include:

- Daily session trends by channel
- Channel efficiency scatter plot (CVR vs revenue)
- Revenue share donut chart
- Revenue per session comparison
- AI-generated recommendation card

The AI recommendation card synthesizes channel performance into a prioritized action summary for marketing stakeholders.

Sessions	Total GA4 sessions in the selected period.
Revenue	Total purchase revenue in USD.
CVR	$\text{Conversions} \div \text{Sessions}$. A conversion is any session with at least one purchase — multiple purchases in the same session count as one.
Avg. Engagement Rate	$\text{Engaged Sessions} \div \text{Sessions}$. GA4's engagement flag fires at 10+ seconds, 2+ page views, or a conversion. Not equivalent to Universal Analytics bounce rate.
Revenue per Session	$\text{Total revenue} \div \text{sessions per channel}$. Primary efficiency comparator in the bar chart.

Page 2 — SEO & Landing Page Performance

Business question: Does ranking position drive revenue, and which pages should be prioritized for optimization?

The page is organized around three analytical perspectives:

- **Funnel performance:** Where do users drop off between organic acquisition and conversion?
- **SEO effectiveness:** Do higher-ranking pages generate stronger business outcomes?
- **Landing page prioritization:** Which pages combine high business value with optimization potential?

Key visualizations include:

- Organic conversion funnel
- Ranking bucket performance analysis
- Landing page performance table combining SEO and behavioral metrics

The page is designed to help SEO and content teams connect search visibility with downstream conversion performance and identify high-impact optimization opportunities.

Organic Traffic	Sessions from the organic search channel only.
CVR (Organic)	Conversion rate scoped to organic sessions only.
Sales from Organic	Total revenue attributed to organic sessions.
Conversion Funnel	Sessions → Engaged Sessions → Conversions. Computed over the full 3-month dataset for statistical stability, not affected by the date filter.

Page 3 — Competitive SEO Intelligence

Business question: How does merch.google compare to competitors on organic reach and traffic value?

The page is organized around two analytical perspectives:

- **Competitive visibility:** How large is each domain's organic keyword footprint?
- **Search efficiency:** How effectively does each competitor convert keyword coverage into estimated traffic value?

Key visualizations include:

- Organic keyword footprint comparison
- Competitor benchmarking table with traffic efficiency metrics
- Domain-level SEO visibility indicators

The page supports strategic benchmarking by helping stakeholders distinguish between scale advantages, efficiency differences, and potential market positioning gaps across competitors.

Organic Keywords	Number of keywords for which a domain ranks in the top 100 on Google. The primary gap metric.
Traffic per Keyword	Organic traffic ÷ organic keywords. Measures keyword footprint efficiency independent of footprint size.
Estimated Traffic Value	SEMrush's estimate of the monthly USD value of a domain's organic traffic.
Avg. SEMrush Rank	Domain authority rank — lower is stronger. Shown as a sparkline trend on the scorecard.

AI Recommendation Layer

The designed pipeline queries channel performance metrics from the data warehouse and sends them to the Claude API with a structured prompt, writing the response back to BigQuery where it is surfaced directly inside the dashboard as a recommendation card.

In the current version of the project, a single recommendation is generated against the fixed historical dataset as a demonstration of the intended pattern; in a production setup, this would trigger automatically after each pipeline run, ensuring any stakeholder opening the dashboard sees a current, data-grounded recommendation

6. Data & Methodology Notes

Data Scope & Temporal Context

The GA4 dataset used for this project is a static public export covering November 2020 to January 2021. It was selected because it provides a realistic ecommerce event stream suitable for demonstrating end-to-end BI pipeline design, with analytical patterns and architecture directly transferable to a live client environment. The SEMrush data is a point-in-time snapshot collected via API during a free trial period and is not automatically refreshed.

One important methodological constraint is that the two datasets do not cover the same time period. The GA4 sessions are historical, while the SEMrush snapshot reflects the site's more recent organic presence. Where both sources are joined at the landing page level, the SEMrush attributes should therefore be interpreted as indicators of the site's broader keyword landscape rather than precise matches to the historical traffic period.

Cross-Source URL Normalization

Joining GA4 sessions to SEMrush keyword data required resolving structural differences between the two sources. Three normalization issues were identified and addressed:

- Hostname mismatch between the legacy and current domains
- Path prefixes present in GA4 URLs but absent from the modern site structure
- Legacy file extensions in GA4 category URLs no longer used on the current site

The solution uses a four-step normalization pipeline applied consistently across both the Python ingestion layer and the SQL staging layer: path normalization, regex-based category classification, SQL-side normalization, and a three-way fallback join strategy.

Joinable coverage typically falls within the 60–75% range. Non-joinable rows are retained for competitor benchmarking purposes but excluded from session-level reporting metrics.

One architectural limitation is that the category classification rules are currently maintained independently in both the Python and SQL layers. In a production deployment, these rules would be centralized into a shared configuration source to eliminate synchronization overhead.

7. Scaling to Production

The pipeline was designed with production deployment in mind. The BigQuery warehouse structure, Looker Studio reporting layer, and Claude API integration are all production-ready as currently implemented.

The remaining automation components are the SEMrush ingestion workflow and the AI insights generator. Both scripts are modular, independently schedulable, and designed to run as Cloud Run jobs triggered by Cloud Scheduler. In a production environment, the SEMrush ingestion would run weekly, while the AI insights pipeline would run daily.

The primary implementation change required for full automation is replacing the hardcoded date windows in the channel summary view with rolling 30-day logic, allowing comparison periods to update automatically with each execution.

The append-only design of the AI-generated insights table provides a built-in historical audit trail, with every pipeline run generating a new timestamped recommendation record without requiring additional logging infrastructure.